

Study Guide

Introduction to Human-Computer Interaction | IT1906 | based on Handouts 01 & 02

1. What is HCI?

Human-Computer Interaction (HCI) is the cross-disciplinary study of how humans design, use, and evaluate interactions with computing devices across productivity, safety, and entertainment.

Two core focus areas:

- **High usability** — focuses on system correctness, efficiency, safety, utility, learnability, and retention.
- **User experience (UX)** — focuses on creating satisfying, helpful, aesthetically pleasing, and motivating interactions.

Key Terms to Remember

Interaction = The abstract conceptual model of how a human interacts with a device to complete a given task.

Interface = The physical and software boundary where two independent systems (human and computer) meet and communicate.

2. Why HCI Matters

Good interfaces increase workplace productivity and ease of use. Poor interfaces make simple systems frustrating.

Aesthetic appeal is now a critical business differentiator. Apple products succeed commercially by prioritizing intuitive, attractive interaction design even when hardware capabilities match competitors.

Major historical milestones driven by HCI:

- **Spreadsheet interface** → made personal computers viable for commercial business computing.
- **Web browser interface** → made the Internet universally usable for non-technical users.
- **Touchscreen interfaces** → enabled global adoption of modern smartphones and mobile devices.
- **Motion-sensing interfaces** → introduced natural bodily actions into video games and interactive entertainment.

3. The 7 Principles of HCI

Principle	What It Means	Real-World Example
Know the User	Design around the target users' age, gender, education, and computing skills (User-Centered Design).	Profiling company employees before deploying workplace software.
Understand the Task	Map the sequence and structure of subtasks needed to complete the user's objective.	Structuring a clean multi-step account login or checkout flow.
Reduce Memory Load	Keep short-term memory load light (~5 to 9 items capacity) to prevent cognitive errors.	Providing keyboard shortcuts (Ctrl+C, Ctrl+V) and concise menus.
Strive for Consistency	Use uniform layouts, icons, and terminology across screens to reinforce familiarity.	Shared menu styling and keyboard commands across Microsoft 365 apps.
Remind Users	Provide active system feedback so users know their action was received and processed.	Confirmation notifications for online payments and OS update alerts.
Prevent Errors / Reversal	Design interfaces that block mistakes before they happen and make actions easily reversible.	Confirmation prompts before deletion and the Undo button.
Natural Interaction	Mirror real-life human physical actions and intuitions directly in system controls.	Nintendo Wii motion gestures and Pokémon GO location interactions.

4. Principles vs. Guidelines

Principles are broad, fundamental concepts that apply widely across all user interfaces.

Guidelines are specific, prescriptive recommendations that direct exact implementation details. Guidelines are more detailed than principles, but they often compete or conflict, requiring designers to make deliberate trade-offs.

HCI Guideline Definition

HCI Guideline = A recommended set of specific information and rules used to direct user interface design and software implementation.

5. Guideline Categories (Jennifer Tidwell)

Category	Scope & Description	Examples
General HCI Design	Display layout, information hierarchy, input collection, visual aesthetics	Color balance on websites, icon dimensions
User Type	Age group, accessibility needs, occupation, culture, and gender	Kids, elderly adults, visually impaired users
Platform / System	Form factor, hardware specs, network environment, operating system	Smartphones, self-service kiosks, e-readers
Vendor / Organization	Brand identity guidelines, public vs. private organization standards	NASA, Apple iOS, Google Material, Windows XP
Interface Style / Modality	Interaction mode: WIMP, touch, voice, gestures, or 3D visuals	Voice assistants, multi-touch gestures, GUI widgets
Operational Context	Physical location, environment noise, lighting conditions, body posture	In-car displays, noisy subways, hands-free work
Applications	Functional purpose and software domain	Social media feeds (Facebook), creative software (Photoshop)

6. Specific Design Guidelines

Design Area	Specific Implementation Guidelines
Visual Display Layout	• Keep screens uncluttered and eliminate redundant elements. • Place primary navigational and actionable items consistently at top center. • Structure data for easy visual comparison and establish clear visual hierarchy. • Balance display density, align related elements, and enforce readable line lengths. • Keep critical controls permanently visible through fixed frames.
GUI Data Entry	• Standardize data-entry actions across all input forms. • Minimize mandatory user typing and keystrokes. • Reduce short-term memory load by presenting selectable options instead of recall. • Ensure entered data directly mirrors how information will be displayed. • Label fields and buttons with clear, unambiguous terms. • Keep semantically opposing choices (e.g., Save vs. Delete) clearly separated. • Sequence fields to follow natural reading and physical scanning flows.
Mobile Optimization	• Display instant network connection and background service statuses. • Minimize typing by supporting touch, voice, and intelligent autofill. • Maintain intense task focus to prevent screen clutter in compact views. • Provide large hit targets for accurate fingertip selection and gesture inputs. • Utilize limited screen real estate with high visual efficiency.

Principle	Core Meaning	Key Accessibility Requirements
P — Perceivable	Information must be presentable in ways all users can perceive.	Provide text alternatives for images, captions for video/audio, and strong contrast.
O — Operable	UI elements and site navigation must be fully operable.	Ensure 100% keyboard accessibility, generous reading time, and zero seizure triggers.
U — Understandable	Content and operational mechanics must be clear and predictable.	Keep language readable, interactions predictable, and offer error correction prompts.
R — Robust	Content must withstand technological evolution across user agents.	Maximize compatibility with assistive technologies, screen readers, and future devices.

7. Quick Review Questions

1. What is the fundamental difference between an HCI principle and an HCI guideline?
2. Differentiate the terms 'interaction' and 'interface' in human-computer systems.
3. What are the two core focus areas of HCI, and what specific attributes does each prioritize?
4. Why is human short-term memory load an essential consideration in interface design, and how many items can it hold?
5. What does the acronym POUR stand for under WCAG 2.1 accessibility standards?
6. Name three of Jennifer Tidwell's guideline categories and provide a real-world design example for each.